Distance-Constrained Dynamic Subtraction Games

Theorem A: Interval Structure and Eventual Periodicity

Research Team – draft for independent mathematical review

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Abstract

Proved from the rules of the game. This standalone paper gives definitions, proof logic, precise evidence boundaries, and cited sources in the same document.

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1 Theorem A: complete structural results

Proof status: Proved.

1.1 Rules, notation and outcome recurrence

Fix $k \geq 1$, $m \geq 0$ and a heap of $n \geq 0$ counters. From a state (n, s), where $s \in \{1, \ldots, k\}$ is the preceding removal, a legal move removes r counters such that $1 \leq r \leq \min(k, n)$ and $|r - s| > m$. The resulting state is $(n - r, r)$. With no previous removal, the first move has only the first restriction. Normal play applies: a player with no legal move loses. We call losing and winning states respectively $\mathcal{P}$- and $\mathcal{N}$-positions, following standard combinatorial-game terminology [1, 2, 3].

Define the P-state row and its potential P-reaching removals by

$$Q_n = \{s \in \{1, \ldots, k\} : (n, s) \text{ is } \mathcal{P}\}, \tag{1}$$

$$A_n = \{r : 1 \leq r \leq \min(k, n),\ r \in Q_{n-r}\}. \tag{2}$$

In particular $Q_0 = \{1, \ldots, k\}$, since there is no legal removal. Starting position n (no previous move) is a $\mathcal{P}$-position precisely when $A_n = \varnothing$.

Theorem 1.1 (Exact interval theorem)**.** *For every $n \geq 0$, the set Q_n is empty, full, or one interval of consecutive integers. More exactly, if $A_n = \varnothing$, then $Q_n = \{1, \ldots, k\}$. Otherwise*

$$\boxed{Q_n = [\max\{1, \max A_n - m\},\ \min\{k, \min A_n + m\}] \cap \mathbb{Z}.}$$

An interval whose lower endpoint exceeds its upper endpoint is empty. Any non-full row has at most $2m+1$ *entries.*

Proof. The legal move r takes (n,s) to a losing state exactly when $r \in A_n$ and $|r-s| > m$. Hence (n,s) is losing precisely when *every* move in A_n fails its previous-removal constraint. Equivalently, $|r-s| \le m$ for every $r \in A_n$. If A_n is empty, this holds for all $1 \le s \le k$. Otherwise the inequalities $r-m \le s \le r+m$ for all $r \in A_n$ intersect to $\max A_n - m \le s \le \min A_n + m$, clipped to $1 \le s \le k$. Its length is at most $2m+1$. □

Lemma 1.2 (Finite-memory recurrence)**.** *For fixed* k, m, Q_n *is computed deterministically from* $Q_{n-1}, \ldots, Q_{n-k}$ *whenever* $n \ge k$.

Proof. The membership test $r \in Q_{n-r}$ determines A_n for each $1 \le r \le k$; then Theorem 1.1 determines Q_n. □

Theorem 1.3 (Eventual P/N periodicity)**.** *For every fixed* $k \ge 1$ *and* $m \ge 0$*, the sequence* $(Q_n)_{n \ge 0}$ *is eventually periodic, and so is the unrestricted starting-position P/N sequence.*

Proof. Each Q_n is a subset of $\{1, \ldots, k\}$, so there are at most 2^k row values. Consider consecutive ordered windows $(Q_n, \ldots, Q_{n+k-1})$. At most 2^{k^2} such windows exist. Theorem 1.2 makes the successor window a deterministic function of its predecessor once the recurrence has entered the full k-move range. Some two windows recur by pigeonhole, and all later windows agree by induction. Thus the rows are ultimately periodic. The starting $\mathcal{P}$ predicate $A_n = \varnothing$ depends on the preceding k rows for $n \ge k$, so it inherits eventual periodicity. □

Corollary 1.4 (Specialization $m = 1$)**.** *Every non-full* Q_n *has at most three consecutive elements.*

Scope. Eventual periodicity is not a statement about a least period or an affine formula in k. Those stronger claims require separate work.

Works cited and source verification

References

[1] R. Sprague, "Über mathematische Kampfspiele," *Tôhoku Mathematical Journal*, first series, vol. 41 (1935), pp. 438–444.
Original journal record: https://www.jstage.jst.go.jp/article/tmj1911/41/0/41_0_438/_article. The metadata and page span were checked against the journal listing.

[2] P. M. Grundy, "Mathematics and Games," *Eureka*, no. 2 (1939), pp. 6–8.
Bibliographical entry and link to scanned source at https://oeis.org/A002188/a002188.pdf; the bibliographical description is corroborated by OEIS. The scan was not used for any copied proof.

[3] E. R. Berlekamp, J. H. Conway, and R. K. Guy, *Winning Ways for Your Mathematical Plays*, vol. 1, second ed., A K Peters, 2001.
Publisher record: https://www.routledge.com/Winning-Ways-for-Your-Mathematical-Plays-Volume-1-2-Berlekamp-Conway-Guy/p/book/9781568811307; verified against publisher and author's publication listings.

[4] V. Gurvich, "On Repetition-free Subtraction Games and Vile-dopey Integers," RUTCOR Research Report RRR-01-2012, Rutgers University, January 2012.
Author's publication list: https://rutcor.rutgers.edu/~gurvich/publist.pdf (entry 240). A report transcription was located, but its detailed theorems were *not* invoked here.

Source-use boundary. The P/N and Sprague–Grundy methods are standard background and are credited above. All specific interval, finite-window, template-obstruction and finite-range arguments in these documents are written self-contained from the game definition, not borrowed from an external proof. Originality relative to *all* prior literature has not been established; a thorough review remains necessary.